

AKHIL JAINI

Data Scientist | Machine Learning Engineer

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SUMMARY

Experienced Computational and Data Scientist with hands-on experience designing real-time, high-performance data systems, predictive statistical models, and applied machine learning solutions for large-scale, noisy datasets. Bringing expertise in advanced Python and SQL analytics, and scientific computing. Proven track record of translating complex analytical problems into production-grade software, delivering measurable performance gains, and communicating technical findings to both technical and non-technical stakeholders. Currently completing a PhD focused on real-time anomaly detection and statistical experimentation at extreme data scale.

TECHNICAL SKILLS

Programming: Python, SQL, C, C++, CUDA C, Rust, JavaScript, TypeScript

Machine Learning: PyTorch, Scikit-Learn, TensorFlow, Regression, Random Forest, RNNs, K-Means & Hierarchical Clustering, Statistical Modelling, Hypothesis Testing, A/B Testing, Feature Engineering

LLM & GenAI Systems: LangGraph (Multi-Agent Systems), Retrieval-Augmented Generation (RAG), LLM evaluation (vLLM, Ollama), Parameter-Efficient Fine-Tuning (LoRA), Model Quantisation, Prompt & Context Engineering

Software Engineering: FastAPI, REST APIs, Docker, Git, GitHub Actions, Unit Testing, CI/CD

Data & Infrastructure: PostgreSQL, SQLite, Linux, AWS, Distributed/HPC Computing (OzStar/Slurm), Data Visualisation, EDA

EXPERIENCE

PhD Researcher (Real-Time Data Systems) – Swinburne University of Technology, Melbourne

June 2023–Present

- Engineered predictive feature-matching and spatial indexing algorithms across multi-terabyte datasets in Python and SQL, improving localisation precision by 4x (from ~1 to <0.25 arcseconds) and enabling reliable downstream analysis at a precision not previously achievable.
- Built and deployed a real-time streaming pipeline processing radio telescope sensor data at >20TB/s throughput, implementing anomaly-detection algorithms that flag rare, millisecond-scale events with >99% reliability.
- Developed statistical and time-series models to characterise periodicity and signal structure in noisy, high-dimensional data, directly informing follow-up data-collection strategy with a national research telescope.
- Led a multidisciplinary international project and team to analyse and characterise rare transient signals from multiple astronomical instruments; co-led a parallel multi-telescope data collection campaign to validate findings for polarisation and time-series analyses.
- Co-authored 12+ peer-reviewed publications in high-impact astrophysics journals (3 as first author) and presented findings at 15+ invited talks and international conferences, translating complex results into clear, actionable takeaways for technical and non-technical audiences.
- Collaborated with international, cross-functional technical teams to convert research-grade analysis into production-grade software adopted by multiple external organisations.

Research Software Engineer – Indian Institute of Astrophysics, Bangalore

August 2021–April 2023

- Designed and validated automated data-quality workflows to detect and correct systematic errors in high-precision instrument data, improving measurement reliability across a multi-year dataset.
- Built Python and C++ control interfaces and automated data-extraction pipelines for real-time instrument calibration and control, high-precision spectrophotometric data acquisition, and telemetry monitoring on a state-of-the-art space-based ultraviolet telescope.
- Designed and simulated optical system components for a next-generation imaging instrument, validating performance under rigorous laboratory test conditions.

Computational Research Assistant – Raman Research Institute, Bangalore

January 2018–March 2020

- Designed and implemented a full-scale Python simulation framework modelling multi-variable sensor networks for an orbiting radio interferometer, performing predictive risk analysis and scenario modelling under the rigorous operating conditions of outer space.
- Published findings as first author in a peer-reviewed journal, independently scoping, executing, and communicating a complete technical research project.

Software Engineering Intern – National Centre for Radio Astrophysics, Pune

May 2017–July 2017

- Built a GPU-accelerated (CUDA C) real-time signal-processing pipeline for high-throughput event detection, porting and extending existing code and validating performance against simulated streaming data.

ML Engineering Intern – CSIR-Central Leather Research Institute, Chennai

May 2016–July 2016

- Applied unsupervised machine learning (K-means and hierarchical clustering) and advanced image-processing techniques in MATLAB and C++ to automatically detect surface defects and grade material quality, improving on conventional grading methods.

PROJECTS

Bushfire Early Warning System (*PyTorch, FastAPI, Docker*) – In development

- Building an end-to-end predictive classification and risk forecasting pipeline analysing high-dimensional Sentinel-1 SAR imagery to undertake feature selection, cross-validation, and performance monitoring to detect active fire fronts through cloud and smoke cover, and deploy with an interactive dashboard via FastAPI, containerised with Docker and integrated with GitHub Actions CI/CD.

Agentic Document & Code Review Assistant (*LangGraph, Ollama, SQLite, ChromaDB, Pydantic*) – In development

- Architected an asynchronous multi-agent LLM engine using LangGraph and Ollama (Llama 3.1 and Qwen3-Coder), deploying a supervisor orchestrator with SQLite state persistence to dynamically route code analysis and technical documentation tasks.
- Engineering a production-ready RAG & static analysis pipeline integrating ChromaDB vector search, Python AST parsers, and strict Pydantic schema validation into a Dockerised FastAPI WebSockets backend for real-time streaming execution.

Apex Drift: Neuroevolution-Based Autonomous Racing Platform (*PyTorch, Pygame, Deep Learning, Optimisation*)

- Developed an autonomous driving system using neuroevolution, evolving neural-network populations through selection, crossover and mutation to learn driving behaviours without supervised training data.
- Engineered a physics-based simulation environment with raycast perception, non-linear vehicle dynamics and multi-objective fitness optimisation, supported by automated testing and model versioning for reproducible evaluation.

Event Detection & Key Actor Identification in Multi-Person Sports Videos (*PyTorch, YOLOv8, OpenCV, Gradio, Docker, GCP*)

- Modernised a computer vision pipeline for multi-person sports video, combining YOLOv8 object detection, Hungarian-algorithm multi-object tracking and a dual-branch BiLSTM with temporal attention for event classification and key-actor localisation.
- Achieved 99.09% event-classification accuracy on 110 held-out broadcast clips across basketball, cricket and football, and deployed the end-to-end system as a Dockerised Gradio application on Google Cloud Run with automated testing.

AstroConvert (*Flet, NumPy, SciPy, AstroPy*)

- Designed and developed a cross-platform desktop tool that intelligently detects, validates, and transforms heterogeneous scientific coordinate formats, substantially reducing manual preprocessing overhead through automated input recognition.

Diffusion Model Fine-Tuning (*Stable Diffusion XL, LoRA, Fooocus*)

- Fine-tuned a parameter-efficient LoRA adapter for a Stable Diffusion XL foundation model, curating and preprocessing a custom image dataset with automated and manual captioning to ensure concept fidelity without overfitting.
- Configured network rank, alpha, and optimiser hyperparameters to fine-tune U-Net and text-encoder components, and systematically evaluated intermediate checkpoints against baseline outputs using Fooocus to balance prompt adherence against generative flexibility.

Astronomical Candidate Classification Pipeline (*Scikit-Learn, Regression, XGBoost*)

- Built a synthetic data-generation pipeline producing realistic astronomical radio transient candidate cutouts (point-source signals, narrowband RFI, broadband noise) and engineered image-based features including signal-noise ratio, morphology and dispersion characteristics for model training and evaluation.
- Trained and compared Logistic Regression, Random Forest and XGBoost classifiers to distinguish genuine transient signals from interference, then extended the pipeline with Random Forest regression to estimate signal properties and packaged it as an end-to-end Python tool.

Local LLM Inference Benchmarking (*vLLM, Ollama*)

- Built a benchmarking framework comparing local LLM inference engines (vLLM vs. Ollama) across open-weight models including Llama 3.1, Qwen 3-Coder, Qwen 3.8, Gemma 4, and DeepSeek V4-Flash, measuring time-to-first-token, token throughput, and memory utilisation across varying concurrency and context-window settings.
- Evaluated quantisation trade-offs and dynamic batching strategies to compare local serving configurations.

Full list of Projects: <https://github.com/jainiakhil/>

RESEARCH IMPACT

First author of multiple peer-reviewed publications in top-tier international astrophysics journals, developing scalable computational methods, data processing algorithms, and software adopted across international astronomy collaborations.

Full list of Publications: ui.adsabs.harvard.edu/public-libraries/6AWkqcbzTQWau_usyI5V8Q

EDUCATION

PhD in Computational Astrophysics

Swinburne University of Technology, Melbourne, Australia

2023-Ongoing

Master of Technology in Astronomical Instrumentation

Indian Institute of Astrophysics, Bangalore, India and University of Calcutta, Kolkata, India

2020-2022

Bachelor of Engineering (Honours) in Electrical and Electronics

Birla Institute of Technology and Science – Pilani, Pilani, India

2014-2018